DATE: PAGE:

Combinatorics - 1

The branch of mathematics that deals with the study of principle of counting is known as combinatorics.

Toss or event

An exp which has observed the outcomes is called toss / event.

Rule of Sum.

If T_1 is an event that can be performed in m ways and T_2 is an event that can be performed in n ways then ~~both~~ either T_1 or T_2 can be performed in $m+n$ ways but not both.

Rule of product

If T_1 is an event that can be performed in m ways and T_2 is an event that can be performed in n ways then both T_1 and T_2 can be performed in mn ways.

Note

1. The above two can be extended for any finite number of tosses.

Problems.

1. Consider a problem of electing a representative

DATE: PAGE:

from among 30 boys and 20 girls.

i) In how many a representative can be selected.

ii) In how many ways 2 representatives, 1 from boys and another from girls can be selected.

=> Number of ways to select a representative from boys = 30 = m

Number of ways to select a representative from girls = 20 = n

i) ∴ By the rule of sum, no. of ways to select a representative = m+n

= 30+20 = 50 ways

ii) By the rule of product, no. of ways to select 2 representatives one from boys and another from girls = mn

30 x 20 = 600 ways

2. Manufacturing of license plates consisting of two letters followed by the 4 digits

i) If no letter ~~are~~ or digit are allowed to repeat then how many plates are possible

ii) With repetition of only letters, how many plates are possible. $26 \times 26 \times 10 \times 9 \times 8 \times 7$

iii) With repetition of letters and digits, how many plates are allowed $26 \times 26 \times 10 \times 10 \times 10 \times 10$

iv) if repetitions are allowed, how many plates are only vowels and even digits.

$5 \times 5 \times 5 \times 5 \times 5 \times 5$

02468

DATE: PAGE:

=> i) The no. of possible liscence plates with no letter or digits is allowed to repeat
$26 \times 25 \times 10 \times 9 \times 8 \times 7$ ways i.e., 3276000 ways

ii) The no. of possible liscence plates with rep of only letters : $26 \times 26 \times 10 \times 9 \times 8 \times 7$
= 3407040 ways

iii) The no. of possible liscence plates with repetition of letters and digits = $26 \times 10 \times 26 \times 10 \times 10 \times 10$ = 6760000 ways

iv) Vowels = A E I O U
Digits = 2 4 6 8 0

The no. of possible liscence plates with repetition of only vowels and even digits
$= 5^6 = 15625$ ways.

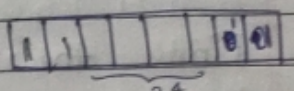
3. A bit is either 0 or 1 and byte is a sequence of 8 bits
find 2^8
i) the no. of bytes
ii) the no. of bytes that begin and end with 1 - 1
iii) the no. of bytes that begin with 11 and not end with 11 2^6
iv) the no. of bytes that begin with 11 or end with 11

$8!$ $^8P_2$ $8 \times 7 = 56$ $\frac{8!}{(8-2)!} = 6!$ $= \frac{8 \times 7 \times 6!}{6!}$

DATE PAGE

i) the total no. of bytes $= 2^8 = 256$

ii) $2^4 = 16$



iii) $2^6 - 2^4$: 2^6 bytes will begin with 11 and 2^4 begins with 11 and end with 11

iv) ∴ the no. of bytes begin with 11 and not end with 11 is $2^6 - 2^4 = 64 - 16 = 48$.

iv) The no. of bytes begin with 11 or end with 11 $= 2^6 + 2^6 - 2^4$ $[\because |A| + |B| + |A \cap B| = |A \cup B|]$

$= 64 + 64 - 16 = 80$.

Permutations

The linear arrangement of distinct objects is called a permutation.

Given a collection of n distinct object of any linear arrangement of r objects can be done in ^nP_r or $P(n,r)$ ways

i.e., $^nP_r = P(n,r) = \frac{n!}{(n-r)!}$

Note

1. $P(n,r)$ or ^nP_r counts linear arrangements in which the objects cannot be repeated
2. If repetitions are allowed, then there are n^r possible arrangements

DATE: PAGE:

Problems

1. Find the no. of votes of 3 distinct letters formed from the letters of the word "JNTU" If repetitions are allowed form the no. of possible 6 letter sequence

=> The no. of votes of 3 distinct letters from the letters of the word "JNTU" is $= {}^4P_3 = \frac{4!}{(4-3)!} = 24$ ways.

If repetitions are allowed the no. of possible 6 letter sequence is $= n^r = 4^6 =$

2. How many different arrangements of the letters in the word "BOUGHT" can be formed if the vowels must be kept next to each other.

=> The given word has vowels O, U. Now treat OU as a single letter. ∴ The total no of letters = 6-2+1 = 5

These 5 letters can be arranged in 5! ways but each of these arrangements, the vowels OU can be arranged among themselves in 2! ways. By the rule of product, the required no. of arrangements = 5! × 2! =

DATE: PAGE:

3 In how many ways can 8 men and 8 women be seated in a row if
i) any person may sit next to any other.
ii) Men and women must occupy alternative seats
iii) generalise the above result for n men and n women.

i) The no. of arrangements of any person may sit to any other = 16!

ii) Men sitting 1st and this can be done in 8! x 8! wdys
Women sitting 1st and this can be done in 8! x 8! ways.
Thus the no of ways men and women must occupy alternative seats = 2 x 8! x 8!

iii) $2(n!)^2$

Arrangements / Permutations with identical objects (eg:- TOO)

If there are n objects of which n_1 are type-1 and n_2 are type-2 so on n_r are type-r with $n_1 + n_2 \cdots n_r = n$ then there are

$$\frac{n!}{n_1!\, n_2! \cdots n_r!}$$

Problems.

1. Find the no. of permutations of the letters of word RECURRENCE

=> There are 10 letters in the given word. Further we have R → 3, E → 3, E → 2, V → 1, N → 1
∴ The total no. of arrangements = $\frac{10!}{3! \times 3! \times 2!}$

2. How many arrangements are there for the letters of the word 'SOCIOLOGICAL'
i) in how many arrangements 'A' and 'G' are together.
ii) in how many arrangements, vowels are adjacent
iii) in how many arrangements, vowels are adjacent and sorted.

=> There are 12 letters in the given word. Further we have, S → 1, O → 3, C → 2, I → 2, L → 2, G → 1, A → 1
∴ The total no. of arrangements = $\frac{12!}{3!\,(2!)^3}$

i) In an arrangement A and G are together as a single letter
Total no. of letters if A and G are adjacent = 12 − 2 + 1 = 11.
∴ No. of arrangements in which A and G are adjacent = $\frac{11!}{3!\,(2!)^3} \times 2!$

ii) The vowels in the given word are A, I, O which are repeated 1, 2, 3 times respectively. We now treat all these vowels namely AIO, IOO in the given word as a single letter. The total no. of letters if all the

DATE: PAGE:

vowels are adjacent = 12 - 6 + 1 = 7

∴ No. of arrangements in which all the vowels are adjacent = $\frac{7!}{2! \times 2!} \times \frac{6!}{5! \times 2!}$

iii) ∴ No. of arrangements in which all the vowels are adjacent and sorted = $\frac{7!}{2! \times 2!}$

3. Find the number of permutations of the letters of the word "INSTITUTION"

i) how many of these are begin with 'I'?

ii) how many of these are begin with I and end with N?

iii) how many arrangements where three Is are together

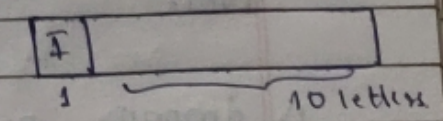
iv) how many arrangements where three Is are not together.

=> Given word has 11 letters

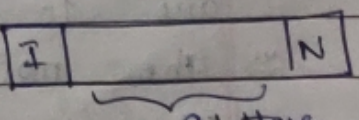
I → 3, N → 2, S → 1, T → 3, U → 1, O → 1.

∴ The no. of arrangements = $\frac{11!}{3! \times 3! \times 2!}$

i) The no. of arrangements begin with I = $\frac{10!}{2! \times 2! \times 3!}$



ii) The no. of arrangements begin with I and end with N = $\frac{9!}{2! \times 3!}$



iii) Consider 3Ts are together as 1 unit, then total no of letters = 11 - 3 + 1 = 9
∴ The no. of arrangements in which 3Ts are together = $\frac{9!}{3!2!}$

iv) The no. of arrangements in which 3Ts are not together = Total no. of arrangements - No. of arrangements in which 3Ts are together
$\Rightarrow \frac{11!}{3!3!2!} - \frac{9!}{3!2!}$

4. How many +ve integers n can be formed using the digits 3, 4, 4, 5, 5, 6, 7 if you want n to exceed 5,000,000?

$\Rightarrow$

Circular Permutations

Arrangements of objects over a closed path is called a circular permutation. The no. of circular permutations of n objects is $(n-1)!$

Note

1) Among n objects, take one of the object and fix the position, then remaining n-1 objects are to be arranged in $(n-1)!$ ways.
2) The no. of C.P of r objects from a collection of n objects is $\frac{1}{r} {}^{n}P_{r}$.

DATE: PAGE:

Problems

1. In how many ways can 10 people be arranged over a circular table. If 3 of them insisting on sitting next to each other, how many arrangements are possible.

=> No. of ways to arrange 10 people over a circular table = $(10-1)! = 9!$
Now consider 3 persons together as one person and these 3 persons can be arranged among themselves in 3! ways.
Total no. of persons = $10 - 3 + 1 = 8$
∴ The no. of ways to arrange 3 among 10 people together over a circular table
$= (8-1)! \times 3!$
$= 7! \times 3!$

Combinations

It mainly deals with selection. The no. of ways in which r objects can be selected from a collection of n distinct objects is denoted by ${}^nC_r$ or $C(n,r)$ or $\binom{n}{r}$ and is given by $\frac{n!}{(n-r)!\,r!}$

Note

1) ${}^nC_r$ is the no. of combinations of r objects from a given collection of n objects. Each combination containing r objects and they can

be arranged among themselves in r! ways
∴ By the rule of product, the no. of permutations = ${}^nP_r$.

2) ${}^nC_r = {}^nC_{n-r}$

3) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

4) In a counting problem, if order isn't imp then we used combinations.

5) In a counting problem, if order is imp then we used Permutation or rule of product.

Problems.

1. A student must answer 5 out of 10 questions in a test including atleast 3 from the last 5 questions. In how many ways, can a student answer 5 questions

⇒ Out of 10 questions, 5 must be answered

First 5 Q	Last 5 Q
$2 \to {}^5C_2$	$3 \to {}^5C_3$
$1 \to {}^5C_1$	$4 \to {}^5C_4$
$0 \to {}^5C_0$	$5 \to {}^5C_5$

∴ No. of ways to answer =
$({}^5C_2 \times {}^5C_3) + ({}^5C_1 \times {}^5C_4) + ({}^5C_0 \times {}^5C_5)$

2. A committee of 8 members is to be formed from 8 teachers and 5 students. In how many ways this can be done if
i) The committee contains exactly two students

DATE: PAGE:

ii) Majority of the teachers
iii) Atleast 2 students
iv) Atmost 6 teachers

No. of teachers = 8	No. of students = 5
i) $6 \rightarrow {}^8C_6$	$2 \rightarrow {}^5C_2$

∴ The no. of ways to form the committee of 8 members with exactly 2 students = ${}^8C_6 \times {}^5C_2$

ii) $8 \rightarrow {}^8C_8$	$0 \rightarrow {}^5C_0$
$7 \rightarrow {}^8C_7$	$1 \rightarrow {}^5C_1$
$6 \rightarrow {}^8C_6$	$2 \rightarrow {}^5C_2$
$5 \rightarrow {}^8C_5$	$3 \rightarrow {}^5C_3$

∴ The no. of ways to form the committee with majority of teachers =
$({}^8C_8 \times {}^5C_0) + ({}^8C_7 \times {}^5C_1) + ({}^8C_6 \times {}^5C_2) + ({}^8C_5 \times {}^5C_3)$

iii) $6 \rightarrow {}^8C_6$	$2 \rightarrow {}^5C_2$
$5 \rightarrow {}^5C_5$	$3 \rightarrow {}^5C_3$
$4 \rightarrow {}^8C_4$	$4 \rightarrow {}^5C_4$
$3 \rightarrow {}^8C_3$	$5 \rightarrow {}^5C_5$

∴ The no. of ways to form the committee with atleast 2 students =
$({}^8C_6 \times {}^5C_2) + ({}^8C_5 \times {}^5C_3) + ({}^8C_4 \times {}^5C_4) + ({}^8C_3 \times {}^5C_5)$

iv) Same as (iii)

3. A committee of 5 people is to chosen from a set of 7 women and 4 men. How many ways are there to form the committee if

$^7C_3 \times {}^4C_2$

i) The committee consists of 3 women and 2 men.

ii) The committee can be any positive size and have equal no. of women and men.

$\Rightarrow ({}^7C_1 \times {}^4C_1) + ({}^7C_2 \times {}^4C_2) + ({}^7C_3 \times {}^4C_3) + ({}^7C_4 \times {}^4C_4)$

iii) The committee has 4 people, and one of them must be Mr. Rama. $\Rightarrow {}^{10}C_3$

iv) The committee has 4 people, atleast 2 are women.

$w = 7$

$2 \rightarrow {}^7C_2$

$3 \rightarrow {}^7C_3$

$4 \rightarrow {}^7C_4$

$m = 4$

$2 \rightarrow {}^4C_2$

$3 \rightarrow {}^4C_1$

$4 \rightarrow {}^4C_0$.

$({}^7C_2 \times {}^4C_2) + ({}^7C_3 \times {}^4C_3) + ({}^7C_4 \times {}^4C_4)$.

v) The committee has 4 people, two of each sex and Mr. and Mrs. Rama cannot both be on the committee.

$\Rightarrow$ case ① = Mrs Rama is on the committee and Mr Rama is not in the committee then one more woman must be chosen from the remaining 6 women and 2 more men must be chosen from remaining 3 men. This can be done in ${}^6C_1 \times {}^3C_2 = 18$ ways.

case ② = Mr. Rama is on the committee and Mrs Rama is not in the committee, then the similar arguement yields ${}^6C_2 \times {}^3C_1$ = 45 ways.

case ③ = Neither of them is on the committee ${}^6C_2 \times {}^3C_2$ ways = 45 ways.

∴ The total no. of ways to form the committee = 18 + 45 + 45 = 108 ways

DATE: PAGE:

4. A committee of 8 is to be formed from 16 men and 10 women. In how many ways.

i) can the committee be formed if there are no restrictions. $= {}^{26}C_8$

ii) there must be 4 men and 4 women.

=> ${}^{16}C_4 \times {}^{10}C_4$

iii) there should be even number of women.

=> $({}^{10}C_0 \times {}^{16}C_8) + ({}^{10}C_2 \times {}^{16}C_6) + ({}^{10}C_4 \times {}^{16}C_4) + ({}^{10}C_6 \times {}^{16}C_2) + ({}^{10}C_8 \times {}^{16}C_0)$.

iv) More women than men.

=> $({}^{16}C_0 \times {}^{10}C_8) + ({}^{16}C_1 \times {}^{10}C_7) + ({}^{16}C_2 \times {}^{10}C_3) + ({}^{16}C_3 \times {}^{10}C_4)$

v) Atleast 6 men.

=> $({}^{16}C_6 \times {}^{10}C_2) + ({}^{16}C_7 \times {}^{10}C_1) + ({}^{16}C_8 \times {}^{10}C_0)$

5. In coding theory a string consists of numbers 0, 1, 2. If $x = x_1, x_2 \ldots x_n$ is one of the string of length n then the weight of x is denoted by wt(x) and is defined as $wt(x) = x_1 + x_2 + \ldots + x_n$. find the no. of strings of length 10 having even weights also find how many strings have 3 zeroes, 4 ones and 3 twos.

=> Let $x = x_1, x_2 \ldots x_{10}$ ① be a string of length 10. Then $wt(x) = x_1 + x_2 + \ldots + x_{10}$.

The string in ① has even weight then there must be an even no. of 1s appeared in the given string.

DATE: PAGE:

No. of 1s	No. of Strings
0	2^{10}
2	$^{10}C_2 \times 2^8$
4	$^{10}C_4 \times 2^6$
6	$^{10}C_6 \times 2^4$
8	$^{10}C_8 \times 2^2$
10	$^{10}C_{10}$

By the rule of sum, the no. of strings of length 10 that have even weight =
$2^{10} + (^{10}C_2 \times 2^8) + (^{10}C_4 \times 2^6) + (^{10}C_8 \times 2^2) \times 10C_{10} \times (^{10}C_2)$

The no. of strings of length 10 will have 3 zeroes, 4 ones and 3 twos $= \frac{10!}{3!\,4!\,3!}$

Note :- In the above problem, how many strings of length 10 will have atleast 8 ones
$\Rightarrow (^{10}C_8 \times 2^2) + (^{10}C_9 \times 2^1) + {}^{10}C_{10}$

6. How many arrangements of letters in MISSISSIPPI have no consecutive S's

$\Rightarrow$ M - 1, I → 4, S → 4, P → 2.

$\boxed{-M-I-I-I-P-P-I-}$

$= 7!$

$\boxed{\frac{11 \times 4! \times 2!}{}}$ → arrangements without S

There are

There are 8 possible places for S's and that can be selected in $8C_4$ ways.

DATE: PAGE:

Thus there are $\frac{7!}{4!\,2!} \times {}^8C_4$ arrangements available with no consecutive Ss.

7. How many arrangements of EFFLORESCENCE has consecutive Cs and consecutive Fs but no consecutive Es

=> E → 4, F → 2, C → 2, L → 1, O → 1, R → 1, S → 1, N → 1

If Cs and Fs are together then we have the following letters, E → 4, F → 1, C → 1, L = O = R = S = N = 1

Without 4 Es, we have the following letters

F = C = L = O = R = S = N = 1

- F - C - L - O - R - S - N -

There are 7! arrangements without Es

There are 8 possible places for E and that can be selected in ${}^8C_4$ ways.

Thus there are $7! \times {}^8C_4$ arrangements with consecutive Cs and consecutive Fs but no consecutive Es.

8. Find the no. of arrangements of all letters in the word TALLAHASSEE. How many of these arrangements has no adjacent As

=> i) $\frac{11!}{1!\,3!\,2!\,1!\,2!\,2!}$

ii) $\frac{9!}{(2!)^3} \times {}^{11-3+1}C_9$

Combinations with repetitions

Suppose we wish to select with repetitions a collection of r objects from a set of n distinct objects. The no. of such selections is given by

$^{n+r-1}C_r = C(n+r-1, r) = \frac{(n+r-1)!}{(n+r-1-r)!\,r!}$

$= \frac{(n+r-1)!}{(n-1)!\,r!}$

$= C(n+r-1, n+1)$

In other words $C(n+r-1, r) = C(r+n-1, n-1)$ represents the no of combinations of n disjoint objects taken r objects at a time with the repetitions allowed.

The following are the other interpretations of this number

* ~~$C(n+r-1, n-1)$~~
* $C(n+r-1, r) = C(r+n-1, n-1)$ represents the no. ways in which r identical objects can be distributed among n distinct containers
* $C(n+r-1, r) = C(r+n-1, n-1)$ represents the no of non-negative integer solutions of the equation $x_1 + x_2 + \dots + x_n = r$, each $x_i \geq 0$, $1 \leq i \leq n$

1. In how many ways can we distribute 10 identical marbles among 6 distinct containers?

$^{n+r-1}C_r = C(n+r-1, r) = \frac{(n+r-1)!}{(n+r-1-r)!\,r!} = \frac{(n+r-1)!}{(n-1)!\,r!} = C(n+r-1, n-1)$.

DATE PAGE

$\Rightarrow$ $n = 6$; $r = 10$.

$C(n+r-1, r) = C(6+10-1, 10)$

$= C(15, 10)$

$= {}^{15}C_{10}$

$= \frac{15!}{(15-10)!\,10!}$

$= 15!/5!10!$

2. Find the no. of non-negative integer solutions of the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 8$.

$\Rightarrow$ Here $n = 5$, $r = 8$. each $x_i \geq 0$, $1 \leq i \leq 5$.

The no of non-negative integer solutions

$= C(n+r-1, r)$

$= C(5+8-1, 8)$

$= C(12, 8)$

$= \frac{12!}{(12-8)!\,8!} = 495$.

3. Find the no of non-negative integer solutions of the inequality $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \leq 10$ where $x_i \geq 0$, $1 \leq i \leq 6$.

$\Rightarrow$ Let $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 10$ where $x_i \geq 0$, $1 \leq i \leq 6$ and $x_7 \geq 1$

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + (x_7 + 1) = 10$ each $x_i \geq 0$, $1 \leq x \leq 7$.

$\therefore x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 9 \rightarrow 1$

here $n = 7$, $r = 9$

DATE: PAGE:

∴ The no of non-negative integer solutions of the eqn = $c(n+r-1, r) = c(7+9-1, 9)$

$= c(15, 9)$

$= \frac{15!}{6!\,9!}$

The above no. represents no of non-negative integer solutions to the ①.

4. Find the no of positive integer solutions of the equation $x_1+x_2+x_3+x_4+x_5 = 30$ where $x_1 \geq 2$, $x_2 \geq 3$, $x_3 \geq 4$, $x_4 \geq 2$, $x_5 \geq 0$

⇒ Let $x_1+x_2+x_3+x_4+x_5 = 30 \rightarrow (1)$ where $x_1 \geq 2$, $x_2 \geq 3$, $x_3 \geq 4$, $x_4 \geq 2$, $x_5 \geq 0$.

$(x_1+2)+(x_2+3)+(x_3+4)+(x_4+2)+x_5 = 30$

$x_1+x_2+x_3+x_4+x_5 = 19 \rightarrow (2)$

each $x_i > 0$, $1 \leq i \leq 5$

∴ The no. of non-negative integer solutions to the eqn (2) is $c(n+r-1, r) = c(5+19-1, 19)$

$= c(23, 19)$

$= \frac{23!}{(23-19)!\,19!}$

$= \frac{23!}{4!\,19!}$

5. A message is made up of 12 different symbol and is to be transmitted through a communication channel. In addn to 12 symbols the transmitter will also send a total of 45 spaces between symbols with atleast 3

spaces between each pair of consecutive symbols. In how many ways can a transmitter send such a message?

=> From 12 symbols we can form 12! messages. For each of these arrangements, there are 11 positions between 12 different symbols. Since there must be three spaces between successive symbols, 33 of the 45 spaces will be used up. The remaining 12 spaces are to be accomodated in 11 positions and this can be done in
$c(n+r-1, r) = c(11+12-1, 12)$
$= c(22, 12)$ ways.
Consequently by the rule of product, the required number $= 12! \times c(22 \times 12)$
$= 12! \times \frac{22!}{10!\,12!}$

$c(11+12-$

$\frac{45}{-33}$
$\frac{}{12}$

$= \frac{221}{10!}$

Binomial Theorem

If x and y are any two variables and n is a positive integer then
$(x+y)^n = \sum_{r=0}^{n} {}^nC_r\, x^r\, y^{n-r}$

Multinomial Theorem

For the positive integers n and k $x_1^{n_1} + x_2^{n_2} + \cdots + x_k^{n_k}$ in the expansion of $[x_1 + x_2 + \cdots + x_k]^n$ is $\frac{n!}{n_1!\, n_2! \cdots n_k!}$ where

each n_i is a non-negative integer less than or equal to n and $[n_1 + n_2 + \dots + n_k] = n$.

Note :-

1) For the positive integers n and k, the coefficient of $x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}$ in the expansion of $[a_1 x_1 + a_2 x_2 + \dots + a_k x_k]^n$ is $\frac{n!}{n_1! n_2! \dots n_k!} \times (a_1^{n_1} \times a_2^{n_1} \times \dots \times a_k^{n_k})$

Problems.

1. Find the coefficient of $x^3 y^9$ in the expansion of $(x+y)^{12}$

$$\Rightarrow (x+y)^n = \sum_{r=0}^{n} {}^nC_r \, x^r y^{n-r}.$$

$$(x+y)^{12} = \sum {}^{12}C_r \, x^r y^{12-r}$$

$$r = 3.$$

$$= {}^{12}C_3 \, x^3 y^9.$$

The coefficient of $x^3 y^9$ in $(x+y)^{12}$ is ${}^{12}C_3$

DATE: PAGE:

2. Find the coefficient of $x^9 y^3$ in the expansion of $(2x-3y)^{12}$

$\Rightarrow (x+y)^n = \sum_{r=0}^{n} {}^nC_r\, x^r y^{n-r}$

$(2x-3y)^{12} = {}^{12}C_r\, (2x)^r (3y)^{12-r}$

$= {}^{12}C_9\, 2^9 x^9 (-3)^3 y^3$

$= {}^{12}C_9$

3. Find the multinomial coefficient of $x^2 y z^3$ in the expansion of $(x+y+z)^6$

$\Rightarrow$ From the multinomial theorem, the coefficient of $x^2 y z^3$ in the given expansion $= \frac{6!}{2!1!3!}$

4. Find the multinomial coefficient of $x^2 y^2 z^3$ in $(3x-2y-4z)^7$

$\Rightarrow$ From the multinomial theorem, the coefficient of $x^2 y^2 z^3$ in the given expansion $= \frac{7!}{2!2!3!} \times 3^2 (-2)^2 (-4)^3$

5. Find the multinomial coefficient of $x^2 y^3 z^4 t^3$ in $(2x-y-z-2t+3)^{16}$

$\Rightarrow$ The general term in the given expansion $= \frac{16!}{n_1! n_2! n_3! n_4! n_5!} \times (2x)^{n_1} (-y)^{n_2} (-z)^{n_3} (-2t)^{n_4} \times 3^{n_5}$

$n_1 = 2,\ n_2 = 3,\ n_3 = 4,\ n_4 = 3$

$n_1 + n_2 + n_3 + n_4 + n_5 = n.$

$2 + 3 + 4 + 3 + n_5 = 16.$

$\boxed{n_5 = 4}$

$$\Rightarrow \frac{16!}{2!\,3!\,4!\,3!\,4!} (2^2)(-1)^3 (-1)^4 (-2)^3 (3)^4$$

6. Find the multinomial coefficient of xyz^{-2} in $[x - 2y + z^{-1}]^4$

$\Rightarrow$ Consider $z^{-1} = t$

$(z^{-1})^2 = t^2$

$z^{-2} = t^2$

Now we need the coefficient xyt^2 in the expansion $(x - 2y + t)^4$

$\therefore$ The coefficient of $xyt^2 = \frac{4!}{1!\,1!\,2!} (1)^1 (-2)^1 (1)^2$

7. Find the multinomial coefficient of $x^2y^3z^{-4}t^3$ in the expansion of $(2x + 3y + 2z^{-1} + t)^{12}$

$\Rightarrow$ Let $p = z^{-1}$

$p^4 = (z^{-1})^4$

$p^4 = z^{-4}$

Ans: $\frac{12!}{2!\,3!\,4!\,3!} \times 2^2 \times 3^3 \times 2^4 \times 1^3$

DATE: PAGE:

Principle of inclusion and exclusion (2 definitions ques)

Consider a set S with $|S| = N$, the condition c_i, $1 \le i \le n$ satisfied by sum of the elements of S.

The no. of elements in S satisfig the condition c_i denoted by $N(c_i)$. $N(c_i, c_j)$ represents the no. of elements in S. They satisfy both c_i and c_j. Ill'y $N(c_1, c_2, \ldots c_n)$ represents the no. of elements in S that satisfy all the conditions.

$N(\bar{c}_i)$ represents the no. of elements in S. They don't satisfy the conditions of c_i.

$N(\bar{c}_i, \bar{c}_j)$ represents the no. of elements in S They don't satisfy both c_i and c_j. Ill'y $N(\bar{c}_1, \bar{c}_2, \bar{c}_n) = \bar{N}$ represents the no. of elements in S. They don't satisfy any of the conditions.

The principle of inclusion and exclusion is given by $\bar{N} = S_0 - S_1 + S_2 - S_3 + \ldots + (-1)^n S_n$ and it represents the no. of elements in S ~~They do not~~ that satisfy none of the conditions $c_1, c_2, \ldots c_n$.

The following expression determines the no. of elements in S, that satisfy exactly m conditions out of n conditions $(0 \le m \le n)$,

$$E_m = S_m - \binom{m+1}{1} S_{m+1} + \binom{m+2}{2} S_{m+2} + \ldots + (-1)^{n-m} \binom{n}{m-n} S_n$$

Further the following expression determines the no. of elements in S that satisfy atleast m conditions out of n conditions $(1 \le m \le n)$

DATE: PAGE:

$$L_m = S_m - \binom{m}{m-1} S_{m+1} + \binom{m+1}{m-1} S_{m+2} - \cdots + (-1)^{n-m} \binom{n-1}{m-1}$$

Note

$N(C_1 \text{ or } C_2 \text{ or } C_3 \cdots C_n)$

$= S_1 - S_2 + S_3 - \cdots + (-1)^{n-1} S_n$

Notes / problems refer pdf due to online class.